5 A small boat is propelled by an engine and initially travels at a constant speed.

The resistive force acting on the boat is directly proportional to the square of its speed.

The power output of the engine is then reduced such that the driving force is reduced to half its original value. After some time, the boat reaches a new constant speed.

What is the ratio $\frac{\text{final power output}}{\text{initial power output}}$?

A 0.125

B 0.354

C 0.500

D 0.707

Commented [YKW(5]: To change phrasing of question

Commented [HWKA(6R5]: updated

Commented [CHN(7]: Consider changing some numbers?

Commented [HWKA(8R7]: done, updated 0.700 to 0.707, which is 1/sqrt(2)